FINAL PROJECT - INSIGHT REPORT

India's workforce and potential exposure to generative AI

A descriptive analysis of occupational change and uneven potential GenAI exposure

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Research question and approach

This report asks: **which Indian workers are concentrated in occupations with greater potential GenAI exposure, and how does this differ by sex, education, and rural or urban location?** It is descriptive, not causal.

The study uses weighted PLFS records for workers aged 15+ in 2018-19, 2021-22, and 2023-24 [1]. It uses nine broad groups because 2018-19 uses NCO-2004 and later waves use NCO-2015.

For 2021-22 and 2023-24, I merge an ISCO-08 task-score proxy from Gmyrek et al. [2]. Compatible four-digit scores form 127 three-digit lower-, middle-, or higher-exposure groups.

33.9% of weighted 2023-24 employment was in agriculture, the largest broad occupation group.	**36.6%** of urban workers were in higher-potential-exposure groups, versus 10.0% of rural workers.	**-0.01** was the Pearson correlation between exposure and 2021-22 to 2023-24 employment-share change.

Employment structure: persistence alongside modest change

Figure 1 is a composition chart: each line shows a broad group's weighted employment share. Agriculture remains largest (33.9% in 2023-24), followed by elementary occupations (22.4%).

From 2021-22 to 2023-24, service and sales gained 2.2 percentage points, craft work 1.3 points, and agriculture 0.9 points.

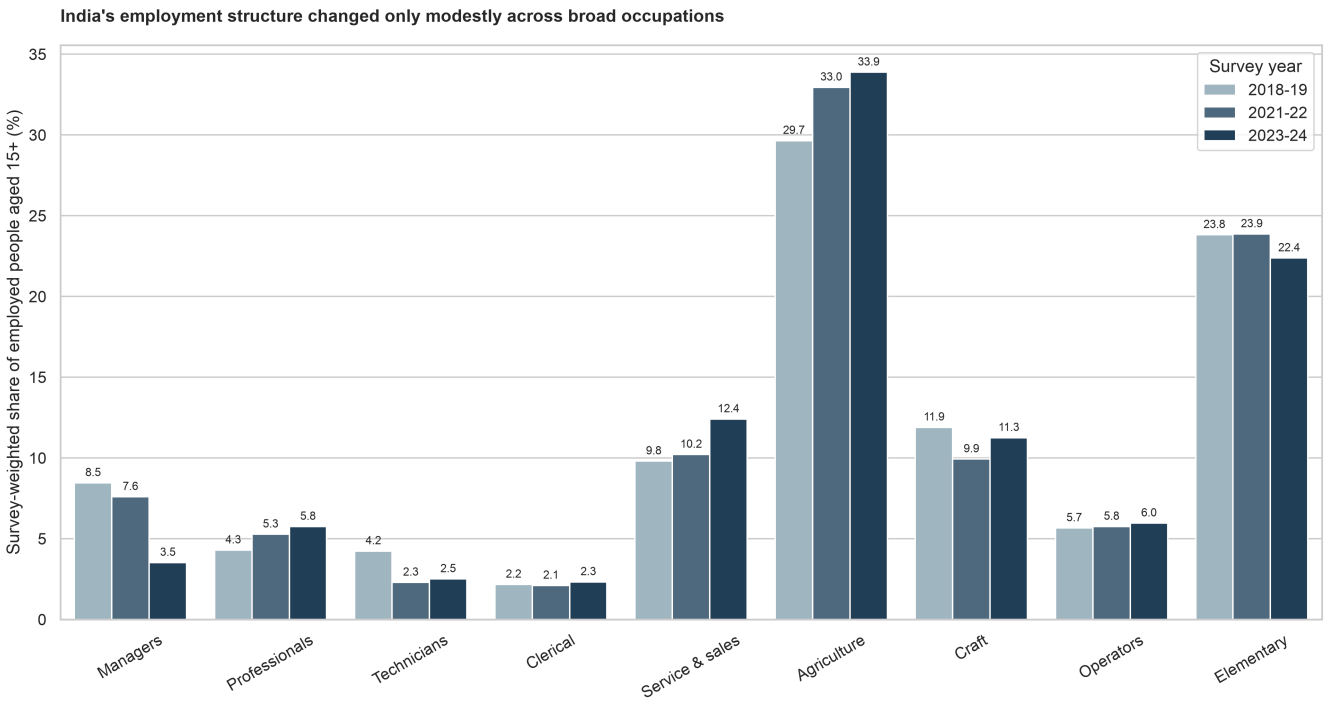


Figure 1. Survey-weighted employment shares by broad occupation. A higher line means that a larger share of employed people were in that group; the 2018-19 comparison is broad because classifications changed.

Where potential exposure is highest

Figure 2 compares detailed groups: longer bars mean higher average potential exposure on a 0-1 scale. Keyboard operators (0.68), numerical/material recording clerks (0.63), and office clerks (0.60) score highest; mining/construction and agricultural labourers score near 0.10. Scores capture tasks GenAI may affect or assist, not job value or job loss.

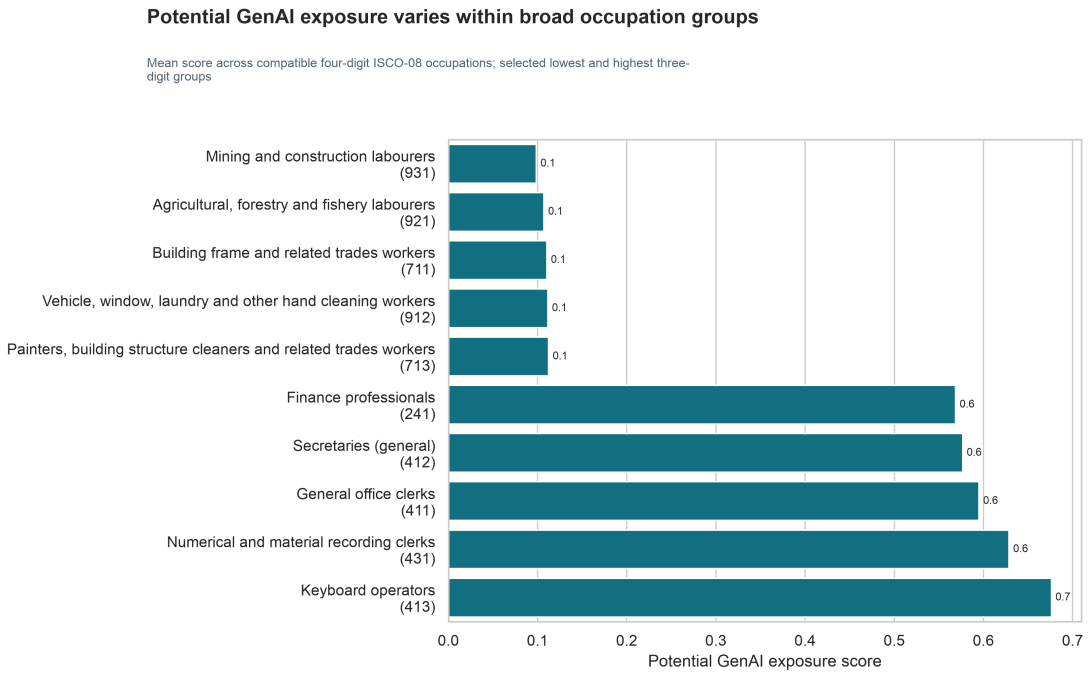


Figure 2. Mean task-level potential GenAI score for selected three-digit ISCO-08 groups. Each bar summarizes compatible four-digit occupations; titles make the coded groups readable.

Exposure is unevenly distributed

Exposure is unevenly distributed: higher-potential occupations include 10.0% of rural workers but 36.6% of urban workers. Shares are 4.3% for below-primary education and 33.3% for secondary education or above; for men and women, 20.0% and 11.0%. These are occupational-composition differences.

Comparison	Share in higher-potential occupations
Rural workers	10.0%
Urban workers	36.6%
Below-primary education	4.3%
Secondary education or above	33.3%
Men	20.0%
Women	11.0%

Recent occupational change and the exposure proxy

Figure 3 shows the overall fitted line plus the three strongest positive within-group fits: service and sales (r=0.46), operators (0.42), and clerical work (0.41). Each dot is a three-digit occupation; its horizontal position is exposure and its vertical position is employment-share change. The black dashed line combines all 127 groups and is nearly flat (overall r=-0.01). Dots above zero gained share. These descriptive correlations are not causal evidence about GenAI.

Other broad-group correlations

Other broad group	n	Pearson r
Elementary	11	0.30
Professionals	27	0.23
Agriculture	9	0.16
Managers	11	-0.10
Technicians	20	-0.16
Craft	14	-0.20

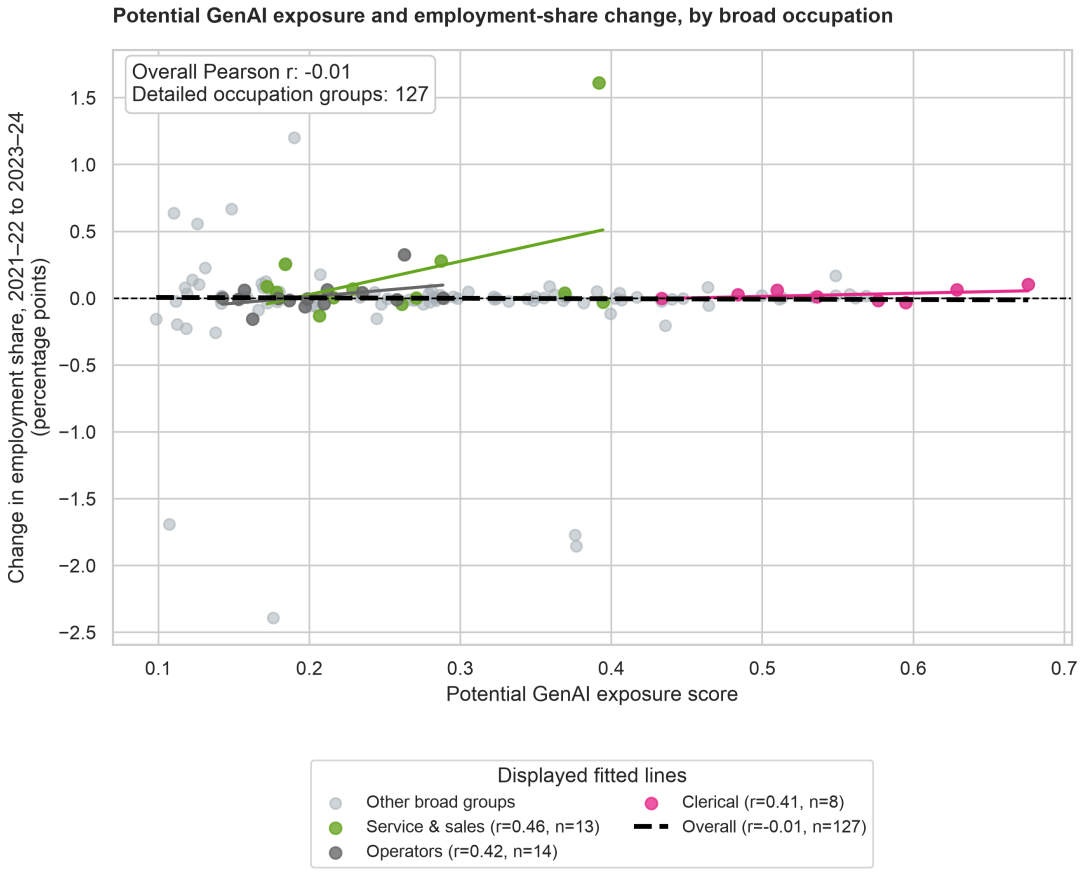


Figure 3. Grey dots are the remaining groups. Colored dots and fitted lines are the three highest positive group correlations; the black dashed line is the overall fit.

Implications, limitations, and recommendations

Potential exposure is not imminent job loss. Urban and more educated workers are more concentrated in occupations where GenAI may complement written and administrative tasks. Training should pair AI literacy with domain skills and include rural and less-educated workers.

Scores are not India-specific adoption measures, and three-digit proxies hide variation within occupations. PLFS waves are cross-sectional, and detailed 2018-19 comparisons are limited by the classification break. Findings are descriptive only.

References

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[2] Gmyrek, P., et al. (2025). *Generative AI and jobs: A refined global index of occupational exposure*. ILO Working Paper 140. Score workbook used: Final_Scores_ISCO08_Gmyrek_et_al_2025.xlsx. https://doi.org/10.54394/HETP0387

[3] Ministry of Labour and Employment, Government of India. (2015). *National Classification of Occupations - 2015*. https://labour.gov.in/sites/default/files/National%20Classification%20of%20Occupations_Vol%20II-B-%202015.pdf

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